Benoît Volant

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PhD in Eco-Design & Decision Science

Work experience

2023-2026 **PhD: *Stakeholders Organization and Environmentally-Oriented Technological Substitution Recommendation: Application to Regional Railway Decarbonization***

Led by Franck Marle, supervised by Yann Leroy, *LGI, CentraleSupelec, University of Paris-Saclay*

- Development of a methodology and a Python model to support decision-making through parametric and prospective life-cycle assessment for the selection of energy technologies
- Development of a methodology to model networks of stakeholders and reduce risks associated with public-private interactions

Papers

- Recommendation through Life Cycle Assessment and Multi-Criteria Decision Analysis: Corsican railway decarbonization, *being published*
- Who decides? Positioning stakeholders in a decision-making process via network modeling and analysis: Application to regional trains decarbonization, *TRIP, March 2026*
- Modeling and Analyzing Interactions Between Stakeholders for Train Decarbonization Decisions at a Regional Scale, *DSM Conference Proceedings, January 2024*

2024 **Feasibility study on local transport services in the Vichy metropolitan area**

- Preliminary study of scenarios meeting the requirements for local transport services

2021 **Project management for the innovative NeoVal rolling stock**

- Track safety compliance assessment and stakeholder management

2020 **Eco-design of an air treatment system and project management**

- Pozzolan air treatment modeling to reduce activated carbon consumption
- Project management for drinking water treatment works

2017-2019 **Identification and digitization of performance indicators for a biogas plant**, *Suez Canada*

- Innovative digester evaluation through indicators identification and analysis

Integration of innovations into the Saint-Étienne (France) drinking water network, *CIRSEE*

- Clean water network analysis and improvement through innovations integration

Study and tools' development for the operation of water treatment works, *Suez International*

- Guide and tool development for reverse osmosis water treatment plants

Classes taught

2026 **Project Management for an eco-design interface**, *CentraleSupelec*

2024-2025 **Lectures on the development and decarbonization of the rail sector**

- *ESTACA, third-year engineering students*
- *Mines de Paris, third-year engineering students*
- *Universidad Politécnica de Madrid, doctoral training*

2023-2026 **Complex system project management for the decarbonisation of the rail sector**, *CentraleSupelec*

2023-2026 **Supervising four trainees on eco-design topics**, *Ikos*

Training

2022 **Summer School of Life Cycle Assessment (LCA)**, *Anne Ventura, EcoSD, Nantes*

2017-2020 **ENGEES: National School of Water and Environmental Engineering**, *Strasbourg*

- Energy recovery and generation
- LCA
- Python modeling

2015-2017 **Mathematics Foundation Course**, *PCSI-PSI, Claude-Bernard, Paris*

Skills

Python, 3 years of experience

LCA tools and databases, Ecoinvent, OpenLCA, BrightWay

English, fluent, TOEIC 985